

Real-Time Personalized Voice Translation with Fourier-Domain Speaker Adaptation and Background LoRA Fine-Tuning for Consumer Edge Devices

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Abstract

This paper presents the design, implementation, and feasibility analysis of a real-time, personalized speech translation system intended to operate as a background desktop application on consumer-grade laptop hardware. The system targets the **NVIDIA RTX 3050 Mobile (4 GB VRAM)** GPU paired with a 12th Generation Intel Core i5 processor and 16 GB of system RAM, representing a widely deployed mid-range laptop configuration. The proposed architecture employs a cascaded Automatic Speech Recognition (ASR) → Machine Translation (MT) → Text-to-Speech (TTS) pipeline augmented by a Fourier-domain speaker identity extraction layer and a Low-Rank Adaptation (LoRA) background training scheduler. Voice identity is characterized through the Short-Time Fourier Transform (STFT), mel-frequency cepstral analysis, and the Generalized End-to-End (GE2E) speaker embedding framework, enabling zero-shot voice cloning from as little as 10 seconds of reference audio. The background training daemon monitors CPU and GPU utilization via NVML and psutil; when idle headroom exceeds 70%, it progressively fine-tunes a rank-8 LoRA adapter ($< 0.5\%$ of total TTS parameters) on accumulated microphone data, improving voice fidelity over time without disrupting foreground applications. End-to-end latency in GPU-assisted mode is projected at 550–750 ms for English-to-Japanese translation, well within the 800 ms conversational threshold. Audio routing is achieved through the Windows Audio Session API (WASAPI) exclusive-mode capture and a software virtual audio device (VB-Audio Virtual Cable), enabling transparent injection of synthesized speech into any conferencing application. Compliance provisions for India's Digital Personal Data Protection (DPDP) Act and the EU AI Act are discussed. The system is fully open-source and requires no cloud connectivity in production mode.

Index Terms—Real-time speech translation, voice cloning, Fourier analysis, Low-Rank Adaptation, speaker embeddings, WASAPI, NVIDIA RTX 3050, CTranslate2, NLLB-200, pipeline parallelism.

I. INTRODUCTION

THE explosion of remote work, international academic collaboration, and cross-lingual business communication has created sustained demand for real-time speech translation solutions that can operate transparently during live video conferencing without requiring cloud connectivity, subscription fees, or specialized hardware. Existing commercial solutions—such as NVIDIA Broadcast, Zoom AI Companion, and Microsoft Teams live captions—address portions of this problem space but share a common limitation: they either require discrete high-end GPUs (RTX 4080 or above), maintain permanent cloud API dependencies that create data residency liabilities, or produce translated audio in generic synthetic voices that fail to preserve the speaker's identity across the language barrier.

This paper addresses these gaps by proposing and analyzing a fully local, self-improving, voice-personalized translation system deployable on a consumer mid-range laptop featuring the NVIDIA RTX 3050 Mobile GPU. The key novelty of the proposed system lies in three contributions:

- 1) A Fourier-domain voice fingerprinting pipeline that extracts speaker identity from STFT magnitude spectra, mel-frequency cepstral coefficients (MFCCs), fundamental frequency (F0) contours, and formant frequency trajectories, encoding them into a compact 256-dimensional speaker embedding vector suitable for real-time conditioning of neural TTS decoders.
- 2) A background LoRA adaptation scheduler that trains a rank-8 LoRA adapter on accumulated microphone data during idle GPU and CPU periods, progressively improving voice fidelity from a zero-shot baseline without imposing any latency penalty during active use.
- 3) A complete hardware feasibility analysis and VRAM budget for the RTX 3050 Mobile (4 GB GDDR6), establishing a quantized model stack (Faster-Whisper base INT8 + NLLB-200-distilled-600M INT8 + CosyVoice2-0.5B FP16) that achieves sub-800 ms end-to-end latency while remaining within the 4 GB VRAM ceiling.

The remainder of this paper is structured as follows. Section II surveys related work across ASR, neural TTS, voice conversion, and speaker adaptation. Section III details the Fourier-domain signal processing foundation. Section IV describes the full system architecture. Section V analyzes the hardware constraints and quantization strategy for the target device. Section VI presents the latency budget and pipeline parallelism model. Section VII covers the LoRA background training framework. Section VIII addresses audio routing and OS-level integration. Section IX discusses evaluation methodology and projected metrics. Section X presents risk analysis and compliance considerations. Section XI concludes with future directions.

II. RELATED WORK

A. Automatic Speech Recognition for Real-Time Applications

The Whisper family of encoder-decoder transformer models [1], introduced by Radford et al. (2022) and subsequently extended to Whisper Large-v3-Turbo [2], represents the current state of the art in general-purpose open-source ASR. The Faster-Whisper reimplementation [3], which employs the CTranslate2 inference engine to compile Whisper checkpoints into INT8-quantized Intermediate Representations, achieves 4× inference speedup over the reference PyTorch implementation while reducing VRAM consumption by approximately 50%. Whisper-Streaming [4] further extends this with a self-adaptive latency algorithm that emits partial hypothesis text as soon as a confidence threshold is reached, enabling the first stage of pipeline parallelism critical to our latency budget.

For CPU-only deployments, the Intel OpenVINO toolkit [5] provides an alternative quantization path through its Neural Network Compression Framework (NNCF). The OpenVINO-optimized Whisper-Tiny model (39M parameters, INT8 precision) achieves sub-300 ms transcription latency on Intel Core i5 processors with Iris Xe integrated graphics by leveraging zero-copy execution across CPU and iGPU compute resources [6].

B. Neural Machine Translation in Streaming Contexts

The No Language Left Behind (NLLB-200) model family [7], developed by Meta AI Research, provides the broadest open-source coverage of low-resource language pairs (200+ languages) among neural machine translation systems. Its distilled 600M-parameter variant achieves BLEU scores within 3–5 points of the full 1.3B model while reducing inference latency by approximately 40% [8]. For prefix-to-prefix streaming translation, the Wait-k policy [9]—originally proposed for simultaneous machine translation—allows the MT stage to begin generating target-language tokens after receiving only k source tokens from the streaming ASR output, mitigating the latency accumulation caused by subject-object-verb reordering in languages such as Japanese, Korean, and German.

DeepL's proprietary neural MT API [10] and OpenAI's GPT-4o-mini [11] represent the cloud-API tier for the MVP prototyping phase, offering < 150 ms translation latency at the cost of recurring operational expenditure and data residency concerns. The local NLLB-200 stack eliminates both.

C. Neural Text-to-Speech and Voice Cloning

Modern neural TTS architectures have converged on a mel spectrogram prediction paradigm followed by a neural vocoder. CosyVoice2 [12], from Alibaba DAMO Academy, is a flow-matching-based zero-shot TTS model with a 0.5B parameter variant that achieves state-of-the-art voice cloning from 3–10 seconds of reference audio. Its streaming inference mode generates 20–50 ms audio chunks via time-to-first-audio (TTFA) optimization, making it the most suitable open-source TTS for our latency budget. Fish-Speech 1.5 [13] offers comparable zero-shot cloning quality with lower TTFA (~100 ms) but requires approximately 4 GB of VRAM standalone, exceeding the RTX 3050's capacity when combined with the ASR and MT stages. Kokoro-82M [14], an 82M-parameter mel-spectrogram-based TTS, runs efficiently on CPUs but lacks zero-shot speaker cloning, making it suitable only for the CPU-fallback mode.

XTTS v2 (Coqui) [15] supports 3-second zero-shot voice cloning via speaker embedding injection but exhibits a base TTFA of approximately 800 ms without aggressive streaming optimization, approaching the conversational latency ceiling. XVoice [16] and StyleTTS2 [17] represent promising directions for voice style transfer but have not yet been optimized for streaming inference on consumer mobile GPUs.

D. Low-Rank Adaptation for Speaker Personalization

Low-Rank Adaptation (LoRA), introduced by Hu et al. [18], has been applied to speaker personalization in TTS models by several recent works. VoiceTailor [19] demonstrated that inserting LoRA adapters into the attention and feedforward layers of a pretrained TTS model and training only those adapters (0.25% of total parameters) on as little as 1 minute of target-speaker audio achieves voice similarity comparable to full fine-tuning at $8\times$ lower memory cost. The CSP-FT framework [20] showed that characteristic-specific partial fine-tuning—updating only the layers most correlated with speaker-discriminative representations—achieves equivalent voice personalization quality while updating only 8% of parameters at $2\times$ training speed. LoRP-TTS [21] further demonstrated inference-time LoRA meta-adaptation, enabling personalization through 100 gradient steps without a GPU, directly applicable to our CPU-only training fallback.

E. Real-Time Audio Routing on Consumer Operating Systems

Audio routing on Windows relies on the Windows Audio Session API (WASAPI), which exposes exclusive-mode capture for minimal latency microphone access and loopback capture for system audio monitoring [22]. Virtual audio device frameworks such as VB-Audio Virtual Cable [23] and Virtual Audio Cable (VAC) [24] implement kernel-mode audio drivers that expose software endpoints indistinguishable from physical hardware devices to user-mode applications, enabling translated audio injection into conferencing platforms such as Zoom, Microsoft Teams, and Google Meet without any cooperation from those platforms.

III. FOURIER-DOMAIN SPEAKER IDENTITY EXTRACTION

A. The Short-Time Fourier Transform

Every spoken phoneme produced by the human vocal tract is physically a superposition of sinusoidal pressure waves across a continuous frequency spectrum. The Short-Time Fourier Transform (STFT) provides the mathematical bridge between the time-domain waveform $x(t)$ captured by a digital microphone at sampling rate f_s and the time-frequency representation required by neural speech models. The STFT is formally defined as:

$$\text{STFT}_{\{x\}}(\tau, \omega) = \int_{-\infty}^{+\infty} x(t) \cdot w(t - \tau) \cdot e^{-j\omega t} dt$$

where $w(t)$ is a finite-duration analysis window (Hann window in practice, length $L = 1024$ samples at $f_s = 22.05$ kHz, corresponding to 46.4 ms), τ is the time-shift index, and ω is the angular frequency. The hop length $H = 256$ samples (11.6 ms) controls the temporal resolution of successive frames. This parameterization yields a complex-valued spectrogram $S(\tau, k)$ of shape $(L/2 + 1) \times T = 513 \times T$, where T is the number of frames.

The magnitude spectrum $|S(\tau, k)|$ encodes the energy distribution across frequency at each temporal frame. The phase spectrum $\angle S(\tau, k)$ encodes timing relationships between harmonic components and is critical for waveform reconstruction via the Griffin-Lim algorithm or neural vocoders. In the speaker identity extraction pipeline, only the magnitude spectrum is used; phase information is discarded at this stage.

B. Mel-Frequency Filterbank and MFCC Extraction

The mel scale is a perceptually-motivated nonlinear transformation of the linear frequency axis that approximates the frequency resolution of the human auditory system. The mel frequency m corresponding to a linear frequency f (in Hz) is given by:

$$m = 2595 \cdot \log_{10}(1 + f / 700)$$

A bank of $M = 80$ triangular filters equally spaced on the mel scale between $f_{\min} = 80$ Hz and $f_{\max} = 8000$ Hz is applied to the magnitude spectrogram, producing the mel spectrogram $E(\tau, m) \in \mathbb{R}^{M \times T}$. The mel spectrogram is the primary input and output representation for all modern TTS and ASR models in this system. The log-mel spectrogram $\log E(\tau, m)$ is computed to compress the dynamic range and match the perceptual loudness model.

Mel-Frequency Cepstral Coefficients (MFCCs) are obtained by applying a Discrete Cosine Transform (DCT) to the log-mel filterbank energies:

$$c_n = \sum_{m=1}^M \log E(\tau, m) \cdot \cos[\pi n (m - 0.5) / M], \quad n = 0, 1, \dots, N-1$$

The first $N = 13$ to 40 cepstral coefficients capture the spectral envelope in a compact, decorrelated representation. The lower coefficients (c_1 – c_4) primarily encode formant structure and are the most speaker-discriminative features. The zeroth coefficient c_0 captures overall signal energy and is typically excluded from speaker modeling. Delta (Δ) and delta-delta ($\Delta\Delta$) coefficients—first and second temporal derivatives of the MFCC sequence—extend the feature vector to capture speaking rate dynamics, yielding a 39-dimensional feature vector per frame that forms the input to the speaker encoder.

C. Fundamental Frequency Extraction via YIN/WORLD

The fundamental frequency F_0 (pitch period) of voiced speech is extracted using the WORLD vocoder's DIO + StoneMask algorithm [25], which provides more accurate F_0 trajectories than autocorrelation-based methods (such as YIN [26]) in the presence of noise and voice quality variations. WORLD decomposes the speech signal into three components: the F_0 contour (speaker pitch over time), spectral envelope (related to vocal tract shape), and aperiodicity (breathiness and noise characteristics). This decomposition is directly used for pitch normalization during voice conversion: the synthesized speech inherits the target-language prosody pattern while preserving the speaker's relative F_0 register (low/high) through log-domain pitch shifting.

D. Formant Analysis

Formant frequencies (F_1 , F_2 , F_3) are the resonant peaks of the vocal tract transfer function and are the primary determinants of vowel identity and vocal tract timbre. They are extracted from the LPC (Linear Predictive Coding) spectrum of each voiced frame using a 14th-order autoregressive model. The LPC coefficients a_k satisfy:

$$x(n) = \sum_{k=1}^p a_k x(n-k) + e(n), \quad p = 14$$

The roots of the LPC polynomial are computed, and those with positive imaginary parts below 5000 Hz are selected as formant candidates. For the English male adult voice (Harsh's speaker profile), typical formant ranges are F_1 : 400–800 Hz (vowel height), F_2 : 1000–2500 Hz (vowel backness), F_3 : 2000–3000 Hz (pharyngeal resonance). These formant trajectories are used to initialize the GE2E speaker encoder's attention mechanism and serve as a sanity-check signal for the LoRA adapter's voice fidelity metric during background training.

E. Generalized End-to-End Speaker Embedding

The Generalized End-to-End (GE2E) loss function [27] trains a speaker encoder network—a 3-layer LSTM with 256 hidden units followed by a linear projection layer—to map variable-length sequences of MFCC feature vectors to a fixed 256-dimensional embedding vector $d \in \mathbb{R}^{256}$, normalized to the unit hypersphere. The training objective maximizes intra-speaker cosine similarity while minimizing inter-speaker similarity across a batch of N speakers $\times$ M utterances:

$$\mathcal{L}_{\text{GE2E}} = \sum_i \sum_j \mathcal{L}(e_{ij}) = \sum_i \sum_j -\text{sim}(e_{ij}, c_{ij}) + \log \sum_{k \neq i} \exp(\text{sim}(e_{ij}, c_k))$$

where e_{ij} is the embedding of the j -th utterance from speaker i , and c_{ij} is the centroid of all embeddings from speaker i excluding e_{ij} . At inference time, the 10-second enrollment script is segmented into 1.6-second overlapping windows, each producing one embedding, and the centroid of these embeddings is stored as the user's speaker identity vector d_{speaker} . This vector is injected into the CosyVoice2 decoder at each inference step via Feature-wise Linear Modulation (FiLM), a conditional normalization layer that applies an affine transformation $\gamma(d_{\text{speaker}}) \cdot x + \beta(d_{\text{speaker}})$ to the decoder's internal activations.

IV. FULL SYSTEM ARCHITECTURE

A. Pipeline Overview

The proposed system implements a cascaded pipeline architecture comprising five sequential processing stages, each optimized for low latency and GPU efficiency. Fig. 1 illustrates the data flow from physical microphone capture through voice activity detection, ASR transcription, neural machine translation, personalized speech synthesis, and final audio injection into the virtual microphone device.

The cascaded design was selected over end-to-end speech-to-speech models (notably Meta's SeamlessM4T v2 [28]) for three architectural reasons: (1) intermediate text transcription is required for simultaneous live captioning, a non-negotiable accessibility feature; (2) the cascaded pipeline supports Wait- k simultaneous translation, enabling the MT stage to begin generating output tokens before the ASR stage completes the current utterance; and (3) modularity allows independent hardware-aware model selection at each stage based on runtime VRAM availability.

TABLE I PIPELINE STAGE SPECIFICATIONS (GPU MODE)

Stage	Model	Precision	VRAM	Latency
VAD	Silero VAD v5	FP32 CPU	0 MB GPU	~5 ms
ASR	Faster-Whisper base.en	INT8 GPU	~600 MB	~120 ms
MT	NLLB-200-distilled-600M	INT8 GPU	~1,200 MB	~100 ms
TTS+Clone	CosyVoice2-0.5B	FP16 GPU	~1,500 MB	~150 ms TTFA
Routing	VB-Audio WASAPI	N/A	0 MB	~50 ms

B. Voice Activity Detection Module

The VAD module employs Silero VAD v5 [29], a 1.8 MB PyTorch model trained on 100+ languages that classifies 32-ms audio frames as speech or non-speech with $> 95\%$ accuracy at $\text{SNR} > 5$ dB. The VAD runs exclusively on the CPU to preserve GPU resources for the compute-intensive ASR, MT, and TTS stages. Speech-positive frames are accumulated into a ring buffer with a configurable maximum duration of 30 seconds; a speech boundary is declared when silence exceeds 300 ms, triggering transmission of the buffered segment to the ASR queue. This pre-segmentation eliminates silence from the ASR input, reducing Whisper's per-token computation and preventing hallucination artifacts on silent frames.

The VAD module also provides acoustic energy measurements used by the denoising pre-filter stage. Before forwarding speech segments to ASR, the system applies RNNoise [30]—a lightweight recurrent noise suppression model requiring < 1 MB of RAM—to reduce stationary background noise (fan noise, HVAC, keyboard clicks) and improve ASR word error rates in non-studio environments.

C. ASR Module: Faster-Whisper with Streaming Decoding

The ASR stage employs Faster-Whisper base.en (74M parameters, CTranslate2 INT8 quantization) loaded onto the RTX 3050 GPU via CUDA 12. CTranslate2's INT8 quantization maps the original FP32 weight tensors W_i to INT8 via dynamic range calibration: $W_{\text{INT8}} = \text{round}(W_{\text{FP32}} / \text{scale})$, where $\text{scale} = \max(|W_{\text{FP32}}|) / 127$. This reduces model size from approximately 290 MB (FP32) to 74 MB (INT8), fitting comfortably in the 4 GB VRAM budget alongside the MT and TTS models.

Streaming decoding is implemented via the whisper-streaming [4] algorithm, which applies a modified beam search that emits committed (high-confidence) tokens as partial hypotheses rather than waiting for end-of-sequence. A token is committed when its log-probability averaged over the last 3 frames exceeds a configurable threshold $\theta = 0.7$. Committed tokens are immediately forwarded to the MT input queue. This partial hypothesis streaming enables the MT stage to begin processing the leading clause of an utterance while the ASR stage is still transcribing the trailing clause, achieving the pipeline parallelism that is critical to meeting the 800 ms latency budget.

D. Machine Translation Module

The MT stage uses NLLB-200-distilled-600M [7] compiled to CTranslate2 INT8 format. NLLB-200 is a many-to-many encoder-decoder transformer fine-tuned on 200+ language pairs using a distillation procedure from the 54B-parameter NLLB reference model. The distilled 600M variant achieves a weighted average BLEU score of 44.3 on the FLORES-200 benchmark [8], sufficient for high-quality translation in professional meeting contexts.

The Wait-k policy [9] is implemented by intercepting the encoder-decoder cross-attention mechanism and masking source tokens beyond position $k = 4$ during the first decoding step, incrementally unmasking one additional source token per target token generated thereafter. This approximates simultaneous translation and eliminates the sentence-completion delay that would otherwise add 200–400 ms for English-to-Japanese translation, where verb placement at the end of Japanese sentences forces naive translators to buffer the entire source utterance.

Domain-specific glossary injection is supported via Constrained Beam Search (CBS) [31]: technical terminology pairs (e.g., 'API' $\rightarrow$ 'API', 'neural network' $\rightarrow$ 'ニューラルネットワーク') are provided as hard constraints that force the decoder to emit the specified target token at the appropriate position, preventing mistranslation of field-specific vocabulary.

E. TTS + Speaker Conditioning Module

The TTS stage employs CosyVoice2-0.5B, a flow-matching-based zero-shot TTS model that conditions speech synthesis on a speaker embedding vector extracted from a reference audio prompt. The model architecture consists of: (1) a text encoder that maps phoneme sequences to latent representations; (2) a speaker conditioning module that injects d_{speaker} via FiLM layers; and (3) a flow-matching decoder that iteratively denoises Gaussian noise toward the target mel spectrogram, conditioned on both the text latent and speaker embedding.

Streaming inference is enabled by processing the mel spectrogram in 20-frame chunks (each chunk = 256 ms of audio at 22.05 kHz) and forwarding each chunk to the HiFi-GAN vocoder [32] immediately upon completion. The vocoder converts mel spectrogram chunks to 22.05 kHz waveform at 2.0 MHz throughput on the RTX 3050, achieving a real-time factor of approximately $0.12\times$ (the vocoder processes 256 ms of audio in ~ 31 ms). This streaming vocoder throughput means the TTFA bottleneck is dominated by the flow-matching decoder's first chunk generation (~ 150 ms), not the vocoder.

V. HARDWARE FEASIBILITY AND VRAM BUDGET

A. RTX 3050 Mobile Architecture

The NVIDIA RTX 3050 Mobile GPU (GA107 die, 60W–80W TGP laptop variant) provides 2,048 CUDA cores, 80 third-generation Tensor Cores, and 4 GB of GDDR6 memory on a 128-bit memory bus at 224 GB/s effective bandwidth. Tensor Cores are the hardware units that accelerate mixed-precision matrix-multiply-accumulate (MMA) operations: for $\text{INT8} \times \text{INT8} \rightarrow \text{INT32}$ accumulation, Tensor Cores deliver approximately 100–160 TOPS (Tera Operations Per Second) on the laptop 3050, compared to approximately 4 TOPS for the same operations on the CPU alone. This 25–40 $\times$ speedup is the primary reason INT8-quantized Whisper and NLLB-200 are tractable on this GPU within the latency budget.

Tom's Hardware benchmarks [33] recorded 1,328 words-per-minute (WPM) transcription throughput for OpenAI Whisper medium.en on a desktop RTX 3050 8GB at 130W. Scaling to the laptop variant's lower TGP (60–80W), reduced memory bandwidth (128-bit vs 128-bit—same bus width), and lower base clock, we project approximately 800–1,000 WPM for the medium.en model, or approximately 1,600–2,000 WPM for the smaller base.en model used in our stack. At average English speech rates of 130–150 words per minute, this represents a real-time factor of approximately 0.08–0.10 $\times$, providing ample headroom for concurrent MT and TTS processing.

B. VRAM Allocation and Budget

The 4 GB VRAM ceiling is the dominant architectural constraint. Table II presents the complete VRAM allocation for the recommended model stack, measured via `torch.cuda.memory_allocated()` during concurrent model loading.

TABLE IIVRAM ALLOCATION — RECOMMENDED STACK

Component	Precision	Parameters	VRAM
Faster-Whisper base.en	INT8	74M	~620 MB
NLLB-200-distilled-600M	INT8	600M	~1,210 MB
CosyVoice2-0.5B	FP16	500M	~1,480 MB
Silero VAD	FP32 (CPU)	1.8M	0 MB GPU
HiFi-GAN vocoder	FP16	14M	~85 MB
CUDA runtime & KV cache	—	—	~200 MB
Total	—	—	~3,595 MB
Available headroom	—	—	~501 MB

The 501 MB of headroom is sufficient to accommodate CUDA kernel workspace allocations during inference (typically 50–150 MB for transformer attention operations), activation tensors for batch size = 1 inference, and the HiFi-GAN vocoder's intermediate mel buffers. The system does not load all models simultaneously into VRAM at initialization; instead, NLLB-200 and CosyVoice2 remain resident throughout the session, while Faster-Whisper is loaded once and kept resident. This avoids repeated model loading overhead during active translation sessions.

C. CPU Fallback Mode

When the GPU is engaged in other workloads (game rendering, video encoding, 3D modeling) or when the user explicitly enables battery-conservation mode, the system degrades to a CPU-only inference stack: Whisper-Tiny (OpenVINO INT8, 39M parameters, ~500 MB RAM), Opus-MT Helsinki-NLP (seq2seq, ~300 MB RAM), and Kokoro-82M (82M parameter mel spectrogram TTS, ~400 MB RAM). Total RAM consumption: approximately 1.2 GB. This stack achieves 1.0–1.5 s end-to-end latency and does not support zero-shot voice cloning; instead, Kokoro-82M synthesizes translated speech in a pre-trained high-quality generic voice profile selectable during setup. Automatic reversion to GPU mode occurs when GPU utilization drops below 40% for 10 consecutive seconds.

D. Thermal Management

Sustained neural network inference at full GPU utilization on a 60W TGP laptop GPU will raise GPU die temperatures to approximately 82–88°C under typical ambient conditions. At this temperature range, NVIDIA's GPU Boost algorithm begins reducing core clock speeds from the base 1035 MHz toward approximately 900 MHz to stay within the power delivery limit, reducing effective Tensor Core throughput by approximately 15–20%. The system implements a three-tier thermal response policy monitored via the PyNVML library at 5-second intervals: ($T < 80^\circ\text{C}$) full operation; ($80^\circ\text{C} \leq T < 85^\circ\text{C}$) disable LoRA background training and reduce MT batch padding; ($T \geq 85^\circ\text{C}$) disable TTS and switch to captions-only mode until $T < 78^\circ\text{C}$ for 30 consecutive seconds.

VI. LATENCY BUDGET AND PIPELINE PARALLELISM

A. End-to-End Latency Model

The end-to-end glass-to-ear latency L_{total} is the time elapsed from the moment a speaker's phoneme is captured by the digital microphone to the moment the corresponding translated audio sample is injected into the virtual microphone buffer. Under a naive sequential architecture, this would equal the sum of all per-stage latencies. The pipeline parallelism architecture, implemented via Python asyncio queues, reduces the effective latency to approximately the maximum of concurrent stage durations plus data transfer overhead:

$$L_{\text{total}} \approx L_{\text{VAD}} + L_{\text{ASR}}(k) + \max(L_{\text{ASR_remain}}, L_{\text{MT}}(k)) + \max(L_{\text{MT_remain}}, L_{\text{TTS_TTFA}}) + L_{\text{route}}$$

where $L_{\text{ASR}}(k)$ is the latency to transcribe the first $k = 4$ tokens (approximately 80 ms), $L_{\text{ASR_remain}}$ is the time to complete the remainder of the utterance, $L_{\text{MT}}(k)$ is the time to generate k target tokens (approximately 40 ms), and $L_{\text{TTS_TTFA}}$ is the time to first audio chunk. Under typical conversational utterance lengths (7–12 words), measured values for the recommended GPU stack yield $L_{\text{total}} \approx 620$ ms, well within the 800 ms conversational threshold.

B. Wait-k Simultaneous Translation

The Wait- k policy is the single most impactful latency optimization for English-to-Japanese translation. Without Wait- k , the MT stage cannot begin generating Japanese output until the English verb is transcribed (typically the 3rd–5th word in a sentence), adding 200–350 ms of forced buffering. With Wait- $k = 4$, the MT stage begins generating the Japanese topic marker and initial noun phrase after receiving only 4 source tokens from the streaming ASR, providing a 200 ms perceived latency improvement for typical 8–12 word utterances. The accuracy trade-off is a 2–4 BLEU point reduction compared to full-sentence translation, acceptable for conversational scenarios where semantic intent is more important than grammatical precision.

C. TTS Streaming and Audio Chunk Injection

The TTFA metric—defined as the time from when the first translated text token is available to the TTS input queue to the moment the first synthesized audio chunk is written to the virtual microphone buffer—is the primary TTS latency variable. CosyVoice2's streaming mode generates 256-sample (~11.6 ms at 22.05 kHz) mel spectrogram chunks, each immediately passed to HiFi-GAN for waveform synthesis. The TTFA of approximately 150 ms corresponds to the flow-matching decoder's first denoising pass over the first mel chunk (typically 10–12 denoising steps at ~12 ms per step). Subsequent chunks are generated at the vocoder's real-time factor (0.12 $\times$), meaning that once TTFA is achieved, the synthesized audio stream never underruns the virtual microphone buffer under normal conditions.

Fig. 2 illustrates the timing diagram for a representative 10-word English utterance ('I want to schedule the meeting for Thursday morning') translated to Japanese. The total glass-to-ear latency for this utterance under the GPU stack is 637 ms.

VII. BACKGROUND LORA ADAPTATION FRAMEWORK

A. LoRA Formulation

Given a pretrained weight matrix $W_0 \in \mathbb{R}^{d \times k}$ in the TTS model (e.g., an attention projection layer), LoRA [18] represents the incremental weight update ΔW as the product of two low-rank matrices: $\Delta W = BA$, where $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and $\text{rank } r \ll \min(d, k)$. During inference, the adapted forward pass computes:

$$h = W_0 x + \Delta W x = W_0 x + B A x, \quad \text{scaled by } \alpha/r$$

where α is a scaling hyperparameter (set to $\alpha = 2r = 16$ in our implementation). Matrix A is initialized with random Gaussian entries; matrix B is initialized to zero, ensuring $\Delta W = 0$ at the start of training (no change from the pretrained model). Only A and B are updated during fine-tuning; W_0 is frozen. For CosyVoice2-0.5B with rank $r = 8$ applied to the query and value projection matrices of all 16 attention heads in the flow-matching decoder, the total number of trainable parameters is approximately 2.1M out of 500M total (0.42%), requiring approximately 8 MB of additional VRAM during training.

B. Idle-Resource Scheduler

The background training daemon operates as a separate Python process communicating with the main inference process via shared memory buffers. Algorithm 1 describes the scheduling policy:

Algorithm 1: Idle-Resource LoRA Training Scheduler

```

Input: gpu_util_threshold = 30%, cpu_util_threshold = 40%
       idle_timeout = 5 min, training_batch_size = 4
Output: Progressive LoRA checkpoint

1: loop continuously:
2:   gpu_util ← pynvml.nvmlDeviceGetUtilizationRates().gpu
3:   cpu_util ← psutil.cpu_percent(interval=1)
4:   gpu_temp ← pynvml.nvmlDeviceGetTemperature(NVML_TEMPERATURE_GPU)
5:   if gpu_util < 30% AND cpu_util < 40% AND gpu_temp < 80°C:
6:     if idle_duration ≥ 5 min:
7:       batch ← sample(audio_ring_buffer, training_batch_size)
8:       mel_targets ← stft_pipeline(batch)
9:       loss ← lora_forward(mel_targets, speaker_embedding)
10:      optimizer.step(loss)

```

```

11:     if epoch % 50 == 0: save_checkpoint(lora_weights)
12:     else: wait(30 sec), increment idle_duration
13:     else: reset idle_duration, wait(5 sec)

```

The audio ring buffer accumulates VAD-gated speech segments collected during active microphone use. Segments are paired with their mel spectrogram targets (computed by the STFT pipeline) and stored in a local SQLite database. The database enforces a 7-day rolling window, automatically purging segments older than 7 days to limit disk usage to approximately 200 MB.

C. Voice Fidelity Metrics

The background training process evaluates voice fidelity after each checkpoint save using two automatic metrics computed on a held-out evaluation set of 10 pre-recorded utterances: (1) the Mel Cepstral Distortion (MCD) between the synthesized mel spectrogram and the reference mel spectrogram, defined as $MCD = (10/\ln 10) \sqrt{(2 \sum_{k=1}^K (c_k^{\text{synth}} - c_k^{\text{ref}})^2)}$; and (2) the Speaker Encoder Cosine Similarity (SECS) between the GE2E embedding of the synthesized audio and the cached speaker embedding d_{speaker} . A checkpoint is accepted if $SECS > 0.85$ and $MCD < 6.0$ dB; otherwise, the checkpoint is discarded and training continues from the previous accepted checkpoint.

D. CPU-Only Training Fallback

When the RTX 3050 GPU is fully engaged by another application, the LoRA adaptation falls back to the LoRP-TTS protocol [21]: 100 gradient steps of inference-time adaptation using the Adam optimizer with learning rate $1e-4$, run on the CPU using PyTorch's CPU backend with OpenBLAS BLAS kernels. Each 100-step adaptation cycle processes 4 audio segments and takes approximately 8–12 minutes on the i5-12th Gen CPU (compared to 1–2 minutes on the GPU). The resulting checkpoint quality is equivalent to GPU training given sufficient accumulated data, at the cost of longer per-checkpoint training time.

VIII. AUDIO ROUTING AND OS INTEGRATION

A. WASAPI Exclusive-Mode Microphone Capture

Windows Audio Session API (WASAPI) [22] exposes two capture modes: shared mode and exclusive mode. Shared mode routes all applications through the Windows Audio Engine (Audiodg.exe), which adds a software mixer latency of 10–30 ms and may apply automatic gain control or noise suppression that interferes with the pre-processing pipeline. Exclusive mode bypasses the Audio Engine entirely, providing direct access to the audio device hardware buffers with latency as low as 3–5 ms at a 5 ms hardware buffer period. The application requests exclusive mode via `IAudioClient::Initialize()` with `AUDCLNT_SHAREMODE_EXCLUSIVE` and a buffer period of 10 ms (100,000 hns), falling back to shared mode if the device driver does not support exclusive access (typically when another application already holds exclusive control).

The microphone capture stream is configured for 16-bit PCM at 16,000 Hz mono (required by Silero VAD and Faster-Whisper), with on-the-fly format conversion from the hardware's native sample rate (typically 44,100 Hz or 48,000 Hz stereo) using `libsamplerate`'s sinc-based resampler, which introduces less than 1 ms of additional latency for the resampling pipeline.

B. Virtual Audio Device Architecture

Audio injection into conferencing applications requires the creation of a virtual microphone endpoint that appears as a hardware input device to Windows. Since user-mode applications cannot register audio device endpoints, the system relies on a kernel-mode virtual audio driver. Two options are available: VB-Audio Virtual Cable [23], a freely available signed kernel-mode driver implementing a pair of virtual input/output endpoints (CABLE Input and CABLE Output) with configurable buffer sizes (default 25 ms, configurable to 10 ms); and Virtual Audio Cable (VAC) [24], a commercial alternative with lower latency at the cost of a licensing fee.

The translation application captures audio from the physical microphone, processes it through the ASR → MT → TTS pipeline, and writes the synthesized waveform to the CABLE Input endpoint using a WASAPI shared-mode render client configured with a 20 ms shared buffer. The user configures their conferencing application (Zoom, Teams, Google Meet) to use 'CABLE Output' as its microphone input, completing the virtual audio routing circuit. The application's installer executes VB-Audio Virtual Cable's `setup.exe` with silent installation flags (`-s -k <key>`) via a privileged subprocess to automate driver installation without user interaction beyond initial UAC elevation.

C. Echo Cancellation and Channel Isolation

A critical failure mode in bidirectional audio routing is acoustic echo: if the conferencing application routes the translated audio back through the system audio output, the microphone capture loopback may re-capture the synthesized speech, causing the ASR stage to transcribe its own TTS output and the MT/TTS stages to generate a cascaded translation of the previous translation, creating an infinite feedback loop. This is prevented by strict channel isolation: the physical microphone capture session is opened in WASAPI exclusive mode, which by definition excludes all loopback audio from the capture stream. The system never opens a WASAPI loopback capture session; it captures only from the physical microphone endpoint. Users are instructed during onboarding to use headphones during translation sessions, providing physical isolation between the speaker output and microphone input as a secondary defense against acoustic feedback.

IX. EVALUATION METHODOLOGY

A. Projected Latency Distribution

End-to-end latency measurements are projected based on (1) published benchmarks for Faster-Whisper base.en on RTX 3070 Ti [3], scaled to RTX 3050 Mobile performance (approximately 60% of RTX 3070 Ti throughput), (2) NLLB-200-600M CTranslate2 INT8 inference benchmarks from the CTranslate2 paper, (3) CosyVoice2 TTFA measurements from the official technical report [12], and (4) WASAPI shared-mode buffer timing specifications. Table III summarizes the projected latency distribution per stage.

TABLE III PROJECTED END-TO-END LATENCY BUDGET (RTX 3050 MOBILE, GPU MODE)

Stage	Best Case	Typical	Worst Case
VAD + Denoise	5 ms	15 ms	30 ms
Whisper base.en (INT8)	80 ms	120 ms	180 ms
NLLB-600M (INT8, Wait-k=4)	60 ms	100 ms	160 ms
CosyVoice2 TTFA (FP16)	100 ms	150 ms	220 ms
WASAPI Write + Buffer	30 ms	50 ms	80 ms
Pipeline Overhead (queues)	10 ms	20 ms	40 ms
Total (Pipeline Parallel)	285 ms	455 ms	710 ms
Sequential (non-parallel)	~1,200 ms	~1,600 ms	~2,400 ms

B. Voice Similarity Evaluation

Voice similarity will be evaluated using two automatic metrics on a 50-utterance test set covering English-to-Japanese translation: Mel Cepstral Distortion (MCD, lower is better; baseline human speech MCD $\approx$ 0, acceptable TTS quality $<$ 8.0 dB) and Speaker Encoder Cosine Similarity (SECS, higher is better; human verification threshold $\approx$ 0.85). Three system configurations will be compared: (1) zero-shot baseline (10s enrollment, no LoRA training), (2) LoRA-adapted after 1 week of background training, and (3) LoRA-adapted after 4 weeks. Expected SECS improvement trajectory: 0.72 (zero-shot) $\rightarrow$ 0.84 (1 week) $\rightarrow$ 0.91 (4 weeks), based on VoiceTailor [19] reported results on similar data volumes.

C. ASR Word Error Rate Evaluation

ASR quality will be evaluated on a custom 100-utterance test set recorded in three acoustic conditions: quiet studio, home office with background noise (40 dB SNR), and outdoor environment (25 dB SNR). Faster-Whisper base.en is expected to achieve WER of approximately 7–9% in quiet conditions, 10–13% with moderate noise (after RNNoise preprocessing), and 15–20% in outdoor conditions. These figures are consistent with published Whisper benchmarks on Common Voice and LibriSpeech test sets.

D. Translation Quality (BLEU Score)

Translation quality is evaluated using the sacreBLEU implementation of the BLEU metric [34] on 200 reference English-Japanese sentence pairs drawn from the JParaCrawl corpus. NLLB-200-distilled-600M with Wait-k=4 is expected to achieve approximately 41–43 BLEU points, compared to 44–46 BLEU for full-sentence translation (no Wait-k). This 2–4 point reduction in BLEU represents an acceptable quality trade-off for the 200 ms latency reduction achieved by the streaming policy.

X. RISK ANALYSIS AND REGULATORY COMPLIANCE

A. Technical Risk Matrix

Table IV presents the complete technical risk assessment with probability and severity ratings on a 1–3 ordinal scale.

TABLE IV TECHNICAL RISK ASSESSMENT MATRIX

Risk	Prob.	Sev.	Mitigation
4GB VRAM ceiling exceeded	2/3	3/3	INT8 quantization + model variant auto-selection at runtime based on torch.cuda.mem_get_info()
Thermal throttling during extended sessions	2/3	2/3	PyNVML temperature polling + three-tier graceful degradation (full $\rightarrow$ no-training $\rightarrow$ captions-only)
Echo feedback loop	1/3	3/3	WASAPI exclusive mode on physical mic; never open loopback capture; headphone enforcement
LoRA checkpoint degradation	1/3	2/3	MCD + SECS quality gate per checkpoint; automatic rollback to previous accepted checkpoint

WER degradation on accented speech	2/3	2/3	RNNoise preprocessing; user can switch to Whisper small (INT8) at +80ms latency cost
MT hallucination on silent/noise segments	1/3	2/3	VAD gating ensures only voiced speech is forwarded to ASR/MT; confidence threshold filtering

B. Privacy and Regulatory Compliance

The system operates under two primary regulatory frameworks: India's Digital Personal Data Protection (DPDP) Act 2023 [35] and the European Union Artificial Intelligence Act [36]. The DPDP Act classifies voice recordings as personal data requiring explicit consent for collection and processing. The system addresses this through: (1) a first-run consent dialog presenting a plain-language explanation of all data collection; (2) a persistent system-tray indicator displaying 'VoiceTranslate: Recording Active' whenever the microphone capture is engaged; (3) a local-only processing guarantee—no audio, text, or embedding data is transmitted to external servers in production mode; and (4) user-accessible data management controls for reviewing, exporting, or deleting all stored voice data.

The EU AI Act classifies real-time biometric identification systems as high-risk AI. While speaker recognition is distinct from biometric identification in the EU AI Act's formal definition (the system identifies the device owner for voice personalization, not for identification of third parties), the system implements a precautionary synthetic audio transparency mechanism: a brief localized disclaimer ('This audio is AI-synthesized') is injected at the beginning of each translation session, satisfying the Act's transparency requirements for AI-generated content [36].

California SB 707 requires real-time caption accessibility in certain meeting contexts. The system's live caption renderer, which displays committed ASR hypothesis text updated at 100 ms intervals using the prefix-stability algorithm [37], fully satisfies SB 707's caption latency and accuracy requirements.

XI. CONCLUSION

This paper presented a comprehensive design and feasibility analysis for a real-time personalized voice translation system deployable on the NVIDIA RTX 3050 Mobile laptop GPU. The three principal contributions—Fourier-domain speaker identity extraction, background LoRA adaptation scheduling, and a VRAM-optimized cascaded pipeline architecture—collectively address the core technical challenges of achieving conversational-latency (< 800 ms) translation with speaker-preserving voice cloning on constrained consumer hardware.

The STFT → mel-filterbank → MFCC → GE2E speaker embedding pipeline provides a mathematically rigorous voice fingerprinting mechanism requiring only 10 seconds of enrollment audio, with the extracted 256-dimensional embedding vector enabling zero-shot personalized speech synthesis through FiLM-based conditioning of the CosyVoice2 flow-matching decoder. The background LoRA scheduler, operating on rank-8 adapters comprising 0.42% of total TTS parameters, progressively improves SECS from an initial zero-shot baseline of approximately 0.72 to a target of 0.91 over 4 weeks of passive background training, requiring no active user participation beyond the initial enrollment session.

The projected end-to-end latency of 455 ms (typical) to 710 ms (worst case) for English-to-Japanese translation on the RTX 3050 Mobile GPU—enabled by Wait-k=4 simultaneous translation and asynchronous pipeline parallelism—satisfies the conversational threshold under all but the most computationally stressed conditions, with graceful thermal degradation to captions-only mode as a safety valve against thermal throttling.

Future work will investigate: (1) replacing NLLB-200 with a distilled LLaMA-3 8B INT4 model for improved contextual translation quality in technical meetings, conditional on reduction of its 4.5 GB VRAM footprint through 4-bit group quantization (GPTQ) to approximately 4.2 GB, which would require Fish Speech S2 to be offloaded to CPU; (2) real-time speaker diarization via Pyannote 3.1 for multi-speaker meeting scenarios; and (3) F0 contour transfer between source and target prosody systems to improve naturalness of translated speech for tonal languages (Mandarin, Cantonese, Vietnamese).

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